CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 10: Direct and Inverse Variations

Topic-wise short notes with solved examples

Direct and inverse variation, the unitary method, and problems on time and work, pipes and cisterns, and men and days.

1 Direct variation

- Two quantities are in **direct variation** if an increase (or decrease) in one causes a **proportional** increase (or decrease) in the other.
- If x and y vary directly, the ratio $\frac{x}{y}$ is **constant**: $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.
- Examples: number of articles and their cost; distance and time at a fixed speed; work done and wages.

Number of pens (x)	2	5	8	12
Cost in ₹ (y)	30	75	120	180

- In the table $\frac{x}{y} = \frac{1}{15}$ throughout.

Example 1.1 If 15 notebooks cost ₹675, find the cost of 22 notebooks.

SOLUTION

$\frac{15}{675} = \frac{22}{y}$, so $y = \frac{675 \times 22}{15}$

₹990

Example 1.2 A car uses 12 litres of petrol for 180 km. How far can it go on 20 litres?

SOLUTION

$\frac{12}{180} = \frac{20}{d}$

300 km

2 Inverse variation

- Two quantities are in **inverse variation** if an increase in one causes a **proportional decrease** in the other.
- If x and y vary inversely, the product xy is **constant**: $x_1 y_1 = x_2 y_2$.
- Examples: number of workers and days for a job; speed and time for a fixed distance; number of animals and days a stock of food lasts.

Speed in km/h (x)	20	30	40	60
Time in hours (y)	6	4	3	2

- In the table $xy = 120$ throughout.

Example 2.1 12 workers can build a wall in 15 days. How many days will 20 workers take?

SOLUTION

$12 \times 15 = 20 \times d$

9 days

Example 2.2 A car takes 4 hours at 60 km/h. How long will it take at 80 km/h?

SOLUTION

$60 \times 4 = 80 \times t$

3 hours

Example 2.3 A hostel has food for 120 students for 30 days. If 30 more students join, how long will the food last?

SOLUTION

$120 \times 30 = 150 \times d$

24 days

3 Unitary method

- Find the value for **one unit** first, then for the required number of units.
- In direct variation, divide to get one unit and then multiply. In inverse variation, multiply to get one unit and then divide.

Example 3.1 8 men earn ₹4800 in a day. How much do 11 men earn?

SOLUTION

One man earns ₹600.

₹6600

Example 3.2 6 pipes fill a tank in 40 minutes. How long do 8 pipes take?

SOLUTION

One pipe takes $6 \times 40 = 240$ minutes.

30 minutes

4 Time and work

- If a person does a piece of work in n days, the work done in **one day** is $\frac{1}{n}$.
- If the work done in one day is $\frac{1}{n}$, the whole work takes n days.
- If A takes a days and B takes b days, together they do $\frac{1}{a}+\frac{1}{b}$ of the work in a day and finish in $\frac{ab}{a+b}$ days.
- Wages are shared in the ratio of the **work done**.

Example 4.1 A can do a piece of work in 12 days and B in 18 days. How long do they take together?

SOLUTION

One day's work $=\frac{1}{12}+\frac{1}{18}=\frac{5}{36}$

$\frac{36}{5}=7\frac{1}{5}$ **days**

Example 4.2 A and B together finish a job in 6 days. A alone takes 10 days. How long does B alone take?

SOLUTION

B's one day's work $=\frac{1}{6}-\frac{1}{10}=\frac{1}{15}$

15 days

Example 4.3 A can do a job in 20 days and B in 30 days. They work together for 6 days and then A leaves. How long does B take to finish the rest?

SOLUTION

Work done in 6 days $=6\times\left(\frac{1}{20}+\frac{1}{30}\right)=\frac{1}{2}$. Remaining work $=\frac{1}{2}$.

$\frac{1}{2}\times 30=15$ **days**

Example 4.4 A, B and C can do a job in 10, 12 and 15 days. How long do all three take together?

SOLUTION

$\frac{1}{10}+\frac{1}{12}+\frac{1}{15}=\frac{6+5+4}{60}=\frac{1}{4}$

4 days

Example 4.5 A and B do a job for ₹900. A alone can do it in 12 days and B alone in 6 days. How should the money be shared?

SOLUTION

One day's work is in the ratio $\frac{1}{12} : \frac{1}{6} = 1 : 2$.

A ₹300, B ₹600

5 Pipes and cisterns

- A pipe that fills a tank in n hours fills $\frac{1}{n}$ of it in one hour. This is treated as positive work.
- An outlet that empties a tank in m hours empties $\frac{1}{m}$ in one hour. This is treated as **negative** work.

Example 5.1 Two taps fill a tank in 10 hours and 15 hours. How long do they take together?

SOLUTION

$$\frac{1}{10} + \frac{1}{15} = \frac{1}{6}$$

6 hours

Example 5.2 A tap fills a tank in 8 hours and a leak empties it in 12 hours. With both working, how long does the tank take to fill?

SOLUTION

$$\frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

24 hours

6 More than two quantities

- When three quantities are involved (men, hours a day, days), decide for each one whether it varies directly or inversely with the unknown.
- For the same work: men $\times$ days $\times$ hours a day is constant. In general $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$.

Example 6.1 15 men working 8 hours a day finish a job in 12 days. How many days do 20 men take working 6 hours a day?

SOLUTION

$15 \times 8 \times 12 = 20 \times 6 \times d$

12 days

Example 6.2 10 men dig a trench 50 m long in 4 days. How many men are needed to dig 150 m in 6 days?

SOLUTION

$$\frac{10 \times 4}{50} = \frac{m \times 6}{150}$$

20 men

7 Common mistakes

- Using a direct proportion for men and days. More men need fewer days.
- Adding the times of two workers. Add their one-day work instead.
- Forgetting to subtract the work of a leak or outlet.
- Sharing wages in the ratio of the days taken. Use the ratio of work done.
- Not checking units: hours with hours, days with days.

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